Multi Agent Debate Framework using Evidence Disagreement Triggering and Conflict Type Classification

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***Abstract*—Evidence MAD focuses on investigating whether multi-agent debate should be triggered for every question or selectively activated when the agent sources disagree. Here the system retrieves information from multiple sources , extracts claims and uses a Natural Language Inference (NLI) model to classify relationships between the claims as agreement, contradiction or uncertainty. As the system identifies disagreement the classifier selects the correct debate strategy to resolve the conflict.**

For the evaluation 96 questions were evaluated with six system configurations with the traces saved. This helped us in performing a comparison of system behavior. Evidence MAD triggered debate for 22 of 96 questions whereas the Standard-MAD triggered debate for 50 questions. This indicates that the debates were activated less frequently in the case of Evidence MAD. The median latency was 50 seconds for Evidence MAD but was 55 seconds for Standard-MAD. Though these measurements show that debate frequency and execution time changed it doesn't evidently prove that Evidence MAD is more efficient under controlled conditions or conflict trigger mechanism caused the differences.

Several limitations needs to be considered while interpreting the findings. The execution traces do not contain the gold answer labels due to which accuracy cannot be evaluated. The model identifiers may vary across traces and systems

***Index Terms*—large language models, multi-agent systems, retrieval-augmented generation, natural language inference, debate, evidence verification**

I. Introduction

A search engine can provide several answers to the same question. Sometimes the sources agree. Sometimes they do not since they use different basis data, versions, populations, or meanings. A multi agent debate can be useful when those answers disagree, but it also increases model calls and waiting time. Retrieval augmented generation helps a language model to use external information at answer time [1]. Multi agent debate is used by language model agents to critique and revise candidate answers [2]. Evidence MAD puts an evidence check between retrieval and debate. The system has two main choices. First, it uses a gate to check whether the available claims show disagreement. Second, when the gate opens, it assigns conflict type and selects the prompt strategy. The implemented types are factual, temporal, version, contextual, and source-related. Here the main idea is that different disagreements may need different questions from a critic. A date conflict needs dates alignment. A version conflict needs naming the release and a contextual conflict needs checking population or conditions.

This paper interprets what is present in the result traces. The question set contains 96 question IDs. Each of the six systems has one trace for each of those IDs. We do not score answers because the traces does not contain the gold labels needed to do that correctly. The main finding is about routing, not answer quality. Evidence MAD had debate on 22 questions, compared with 50 for Standard MAD. This shows a lower debate rate for the proposed configuration. The results also show long latency tails, failed internal model calls, many insufficient evidence states, and a mixture of model identifiers across systems. Those details are important when reading the comparison. This paper makes three limited contributions. It describes the implemented retrieval, gate, and debate path, gives trace derived counts and plots for six configurations on the same 96 questions and records limits that prevent an answer quality or causal claim. The paper is required to be seen as a transparent report of a prototype run. This is not a benchmark result proving that Evidence MAD is better than the alternatives.

II. Related Work

The literature review classifies prior work into debate protocols, selective debate, and evidence conflict handling. We include all 16 papers listed there. This is a focused project review, not a systematic review of every multi agent or retrieval system.

A. Multi agent debate

ReConcile uses a round table discussion with diverse language models and confidence weighted voting [3]. Du et al. study repeated proposal, critique, and revision to improve reasoning and factuality [2]. DoT uses a tit for tat discussion and a judge that can stop the process when further discussion is not useful [4]. ChatEval applies role based debate to evaluating generated text rather than answering evidence conflicts [5]. Khan et al. study two stronger agents arguing on opposite answers for a weaker judge, and FORD compares debate

structures to study whether different language models can reach on accepted decisions [6], [7].

These papers establish that agent roles, debate structure, and stopping rules are required design choices. They do not, by themselves, show that debate should start only after a contradiction between independently retrieved claims. Our system makes that pre debate decision explicit and also the current experiment does not establish that its decision improves answer quality.

B. Selective debate and adaptive collaboration

DOWN starts debate as per the confidence of an initial answer [8]. Baba and Chudziak also use confidence gating but for argument relation classification [9]. These works are close to Evidence MAD because both use classification to check if every question needs a debate. Their trigger is model confidence on the task, while the Evidence MAD gate is to use relations among retrieved claims. Choi et al. separate the contributions of voting and debate and caution that debate should be evaluated against simpler aggregation strategies [10]. That result supports our choice to treat debate cost and answer quality as separate measurements.

C. Retrieval and conflicting evidence

MADKE adds a shared retrieval knowledge pool to a multi-agent debate and lets agents select knowledge during discussion [11]. MAIN-RAG uses agents to filter and score retrieved documents before generation [12]. MADAM-RAG directly studies retrieval with conflicting evidence and uses agents to discuss ambiguity, misinformation, and noise [13]. Tool MAD assigns different tools to agents and can adapt retrieval as the debate develops [14]. These systems show that retrieval and debate have already been used in several ways. Evidence MAD's design question is simple: whether a comparison of claims should open the debate path before conflict resolution begins. Other work focuses directly on conflict identification and response. DRAGged into Conflicts introduces a taxonomy for conflicts among search results and studies response behavior in a search augmented single model setting [15]. Xu et al. survey knowledge conflicts, including disagreement between context and model memory, between context sources, and within model knowledge [16]. SMADE IE uses sparse, evidence driven debate for zero shot information extraction, where agents may disagree over labels or extracted structures [17]. These works make it clear that neither evidence conflict nor evidence informed debate is new on its own.

D. Positioning of this study

The system tested here combines three already implemented steps: normalize claims from separate retrieval agents, use pairwise NLI plus query aware rules to assign an evidence state, and, on disagreement, classify the conflict and select a type specific debate prompt. It is different from confidence only gating by using retrieved claim relations as the trigger. It is different from document filtering by making the conflict state control whether the debate stage runs. It is different from conflict taxonomies that guide a single answer by passing the type into a multi agent resolution prompt. These are the differences in system design, not proof of priority. The study is an implementation and a first trace based comparison, its current data do not prove that the combined design is more accurate or more useful than the cited systems.

III. System Design

Figure 1 summarizes the implemented control flow and its three response paths.

A. Retrieval and claim preparation

For each query, the system assigns work to retrieval agents and collects the response evidence. The saved run signature has DuckDuckGo, Wikipedia, Wikidata as configured sources. Retrieved snippets are sent through a normalizer that gives short claims and retains source information. A relevance filter can remove results that are not related to the question. The NO RELEVANCE FILTER configuration disables that filtering step. The normalization stage is important because later decisions use the claims it produces, not the full original pages. If the normalizer loses a date, scope, or qualification, the gate cannot recover it. Each trace has the claims, source metadata, evidence state, debate activity, final answer, and per query metrics. This helps in interpreting why the system took a particular path, even when the final answer cannot be scored.

B. Evidence gate

The gate compares available claim pairs in both directions with an NLI model. NLI models estimate whether one sentence supports, contradicts, or is neutral toward another; MNLI is a standard resource for this task [18]. The Transformers checkpoint `ynie/roberta-large-snli_mnli_fever_anli_R1` is used for the implementation. This also combines inference output with query aware logic and assigns an evidence state: AGREEMENT, DISAGREEMENT, UNCERTAIN, or INSUFFICIENT. In the proposed configuration, disagreement is the state that routes the trace to conflict classification and debate. Uncertain and insufficient states do not count as a positive evidence of agreement. This rule makes the gate simple to inspect. It also creates a limitation: a real conflict that is assigned uncertain or insufficient will not reach the debate stage. That is why debate rate alone is not enough to evaluate the gate. A proper test needs independently labelled evidence conflicts and answers, neither of which is available for the 96 traces.

C. Conflict aware debate

When the proposed system detects disagreement, a classifier selects a prompt strategy from factual, temporal, version, contextual, and source related types. The debate uses proposer, critic, reviser, and judge roles. A debate can stop after a critic finds no unresolved objection, when the round limit is reached, or when a stage fails. The trace records the debate rounds and stop reason when available. The six configurations differ in their intended settings. SINGLE LLM produces a direct

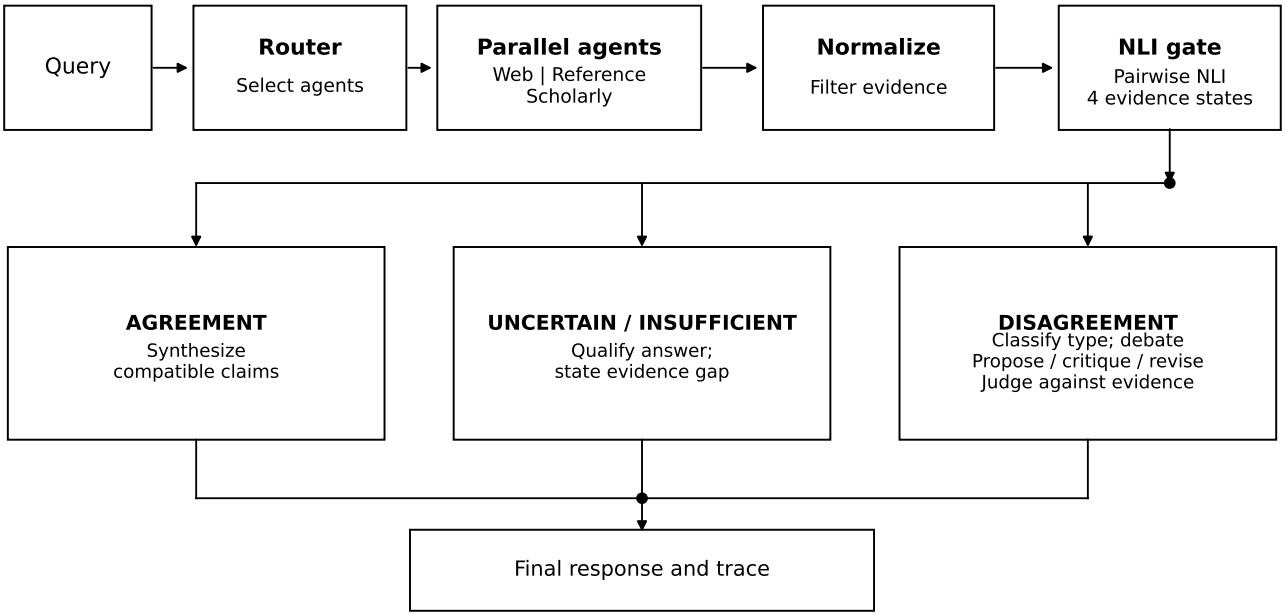


Fig. 1. Implemented Evidence MAD control flow. The four state evidence gate sends disagreement to conflict classification and typed debate. Agreement, uncertain, and insufficient states have separate answer paths. Only the disagreement path enters conflict classification and typed debate.

answer without retrieval based debate. STANDARD MAD uses a generic debate strategy and is configured to debate when there is usable evidence. PROPOSED MAD uses the evidence gate and typed strategies. The three ablations are named NO GATE, NO TYPED STRATEGY, and NO RELEVANCE FILTER. These names describe the intended configuration changes, the values in the trace are used for all reported counts. The observed triggers do not always match the expected configuration description, so we report the outcomes recorded in traces rather than assuming every requested stage completed.

D. Trace interpretation

A trace with `evaluation_status=success` means that the evaluator wrote a completed record. It does not mean that every internal model call succeeded or that the answer is correct. For example, traces can contain a final answer after a proposer call failed. We distinguish completed records from successful internal calls in the results and do not treat trace completion as answer accuracy.

IV. EXPERIMENTAL METHOD

A. Question set

The source dataset contains 96 question IDs identified from the available traces and analysis results. These questions cover everyday factual prompts and specialized topics, as reflected in the recorded queries. The available traces were collected from six system configurations, resulting in 576 trace records across the dataset. However, the available records do not provide usable gold-standard answer labels for each question, and the saved metrics indicate that no key-fact samples were scored.

TABLE I
OPENCODE MODEL IDENTIFIERS RECORDED IN EACH SYSTEM'S 96 TRACES.

System	Cohere alias	OpenRouter free alias
Single LLM	32	64
Standard MAD	88	8
Evidence MAD	91	5
No Gate	86	10
No Typed Strategy	92	4
No Relevance Filter	91	5

Consequently, we do not calculate accuracy, precision, recall, F1-score, missed-conflict rate, or unnecessary-debate rate, as doing so without ground-truth annotations would risk conflating system-generated decisions with verified correctness.

B. Configurations and recorded model metadata

The trace metadata contains multiple OpenCode model identifiers across the evaluated system configurations. Table I summarizes the frequency of each recorded identifier within the corresponding system folders. This variation indicates that the experimental configurations were not uniform with respect to their recorded model identifiers, which may introduce a confounding factor in the comparison. The identifiers `console-openrouter/cohere/north-mini-code` and `console-openrouter/openrouter/free` are retained as recorded metadata; however, the trace signatures alone are insufficient to independently establish the underlying model or provider responsible for each response. Accordingly, these identifiers are reported descriptively, without attributing performance differences to specific models or providers.

The run signature also records `llama-3.3-70b-versatile` as the general LLM setting and the NLI checkpoint described above. The same signature lists the source providers in Section III-A. Because the model identifier mixture differs by system, the timing and routing results are descriptive observations from the saved run, not a controlled comparison that isolates only the system design.

C. Measures

We calculate debate activation as the percentage of traces where the saved `debate_triggered` field is true. Latency uses the per trace `metrics.latency_seconds` field. The median and empirical 95th percentile summarize the distribution. Successful and failed call averages use `total_llm_calls_ok` and `total_llm_calls_failed`. Evidence state counts come from the trace metric field, and retrieval coverage uses the recorded usable agent and claim counts. These are operational measures. They do not establish whether a debate was necessary, whether a conflict was missed, or whether the final response was correct. Those questions need labels created independently of the system outputs. We report no significance tests because the result is a single run with a small matched set and non uniform model metadata.

D. Reproducibility

All figures were generated using Matplotlib from the collected execution traces and their corresponding analysis outputs. The visualizations use a consistent formatting scheme with white backgrounds and black text. Since the experiments involved live web retrieval, search results and network latency may vary across repeated executions. The reported plots therefore represent the observed experimental runs rather than guaranteed identical outcomes under repetition.

V. Results

A. Debate activation

Evidence MAD recorded debate on 22 of 96 questions (22.9%). Standard MAD recorded debate on 50 (52.1%). The No Gate configuration also recorded 50 triggers. No Typed Strategy recorded 23 (24.0%), and No Relevance Filter recorded 18 (18.8%). The single LLM baseline recorded no debate. Figure 2 shows these values. These counts show that the proposed configuration debated less often than Standard MAD in this run. They do not tell us whether the 74 questions without an Evidence MAD debate were appropriate cases to skip. Because the traces does not have conflict labels, we cannot separate correct avoided debates from missed conflicts. The difference is a useful routing observation and a clear target for a labelled follow up evaluation.

B. Whether a triggered debate completed

A debate trigger does not mean a debate finished. In the Evidence MAD traces, 16 of the 22 triggered debates were recorded as completed, five failed, and one was partial. Standard

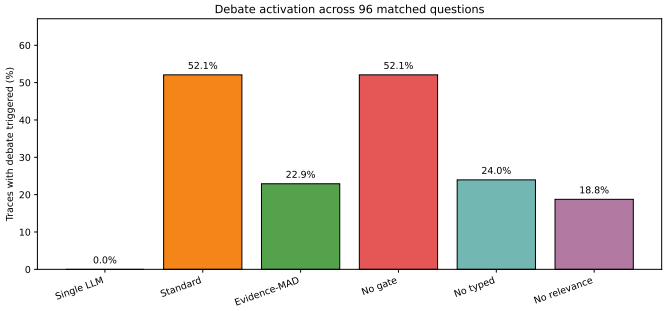


Fig. 2. Fraction of matched traces with debate recorded as triggered. The Single LLM system has no debate stage.

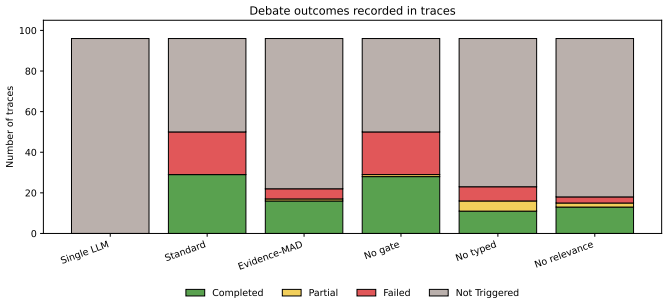


Fig. 3. Recorded debate outcomes. The completed, partial, and failed segments apply to traces with a debate trigger; the remaining segment had no trigger.

MAD recorded 29 completed and 21 failed outcomes among its 50 triggered debates. The no-gate configuration recorded 28 completed, 21 failed, and one partial outcome. These counts come from the debate outcome object, because the separate metrics field does not have a status for every trace. Figure 3 includes both triggered outcomes and questions where debate was not triggered. The failure counts make clear that a lower debate count also changes the number of opportunities for a debate stage to fail. A useful future comparison should report answer quality and completed debate rate together.

C. Evidence states

For Evidence MAD, the trace states were insufficient in 47 cases, uncertain in 26, disagreement in 22, and agreement in one. Only the 22 disagreement traces triggered debate. Standard MAD's traces recorded 46 insufficient states, 29 uncertain states, 16 disagreement states, and five agreement states, while debate was triggered in 50 traces. This gap reflects the way the baseline can force a debate when usable evidence exists; it should not be interpreted as 50 independently verified conflicts. Figure 4 gives counts for all systems. The high number of insufficient and uncertain states is a practical limitation. In 73 of the 96 Evidence MAD traces, the recorded state was not disagreement. Some of these questions may genuinely lack comparable evidence, others may reflect retrieval or normalization problems. The traces alone do not tell us which explanation applies. A future evaluation should label

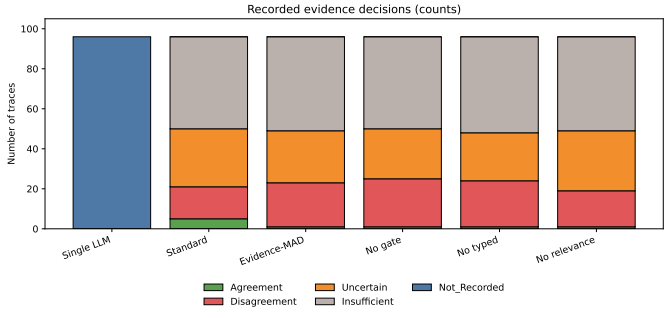


Fig. 4. Evidence-state counts from each system's traces. Single LLM does not record an evidence state.

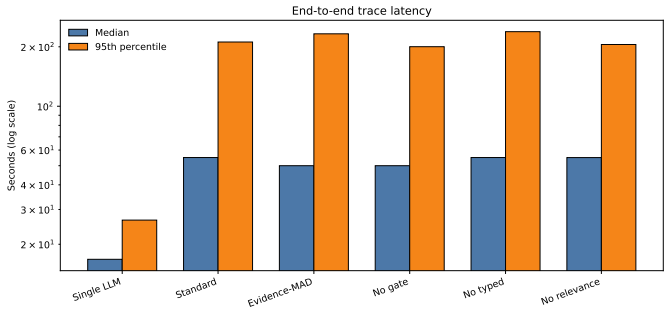


Fig. 5. Median and 95th-percentile latency for the matched traces. The vertical axis is logarithmic.

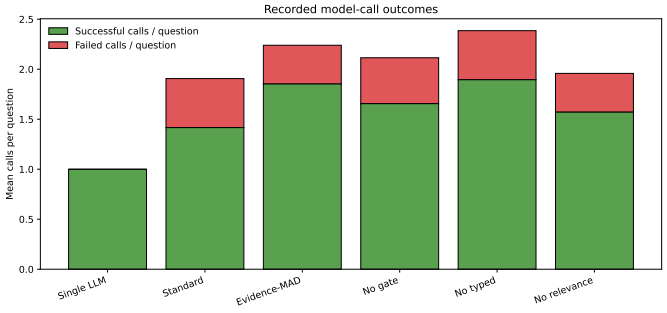


Fig. 6. Mean successful and failed model calls per question, as recorded in each trace.

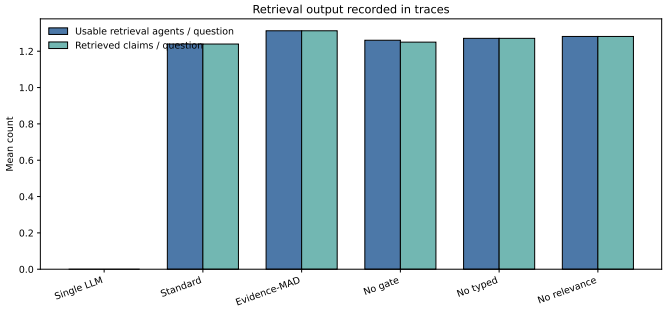


Fig. 7. Mean usable retrieval agents and extracted claims per question.

the retrieved claim pairs and record whether each state is appropriate.

D. Latency and model calls

Evidence MAD had a median latency of 50.0 seconds and a 95th percentile of 232.7 seconds. Standard MAD had a median of 55.0 seconds and a 95th percentile of 211.5 seconds. The medians are close, while the proposed system has a longer observed upper tail. The single LLM baseline was faster in this run, with a 16.8 second median and a 26.5 second 95th percentile. Figure 5 uses a logarithmic axis to show both typical and long tail latency. Evidence MAD averaged 1.85 successful and 0.39 failed model calls per question. Standard MAD averaged 1.42 successful and 0.49 failed calls. The no-typed-strategy ablation averaged 1.90 successful and 0.49 failed calls, the highest successful call mean among the retrieval systems. These averages include calls outside the debate roles, such as normalization. They should not be read as a debate-only cost. Figure 6 displays the saved call counts.

E. Retrieval output and model metadata

The retrieval systems recorded about 1.24–1.31 usable agents per question on average. Evidence MAD recorded 1.31, and Standard MAD recorded 1.24. This mean is low relative to the configured set of retrieval sources. It indicates that many traces had only a small amount of usable evidence available to the next stage. The call to source count is not one to one: an agent may use multiple sources, and a source may produce several snippets. Figure 7 shows usable agents and claims recorded in the traces. Figure 8 shows the model identifiers in the trace signatures. The single LLM traces contains 64 traces with the OpenRouter free alias and 32 with the Cohere alias. The other systems are mostly Cohere alias traces, with between four and ten OpenRouter alias traces each. This imbalance is large enough that it limits direct system to system interpretation. These labels come from the saved signature and are not an independent audit of the API route used at call time.

VI. Discussion

The traces support a narrow conclusion. In this saved run, the proposed configuration triggered debate less often than Standard MAD, and its median latency was about five seconds lower. Its 95th percentile latency was about 21 seconds higher. Without gold labels, we cannot tell whether the lower debate rate shows better selectivity or missed conflicts. Without uniform model metadata, we also cannot attribute the latency differences to the gate or conflict strategy. The ablation results do not form a clean component by component ranking. The no-typed-strategy variant triggered debate on 23 questions, close to Evidence MAD's 22. The no relevance filter variant triggered on 18, fewer than Evidence MAD. These are trace counts, not evidence that either ablation makes better decisions. Retrieval and normalization can change the claims presented to the gate, and those changes can alter both state and routing.

The many insufficient states deserve attention. Evidence MAD had 47 insufficient traces, almost half of the matched set. The system may be appropriately withholding a decision

TABLE II
TRACE-DERIVED OPERATIONAL RESULTS OVER 96 MATCHED QUESTIONS. LATENCY IS IN SECONDS; CALL COUNTS ARE MEANS PER QUESTION.

System	Debates	Debate rate	Latency median	Latency $p95$	Calls OK	Calls failed
Single LLM	0/96	0.0%	16.79	26.51	1.00	0.00
Standard MAD	50/96	52.1%	54.99	211.47	1.42	0.49
Evidence MAD	22/96	22.9%	49.99	232.65	1.85	0.39
No Gate	50/96	52.1%	49.99	200.21	1.66	0.46
No Typed Strategy	23/96	24.0%	54.99	238.55	1.90	0.49
No Relevance Filter	18/96	18.8%	54.96	205.42	1.57	0.39

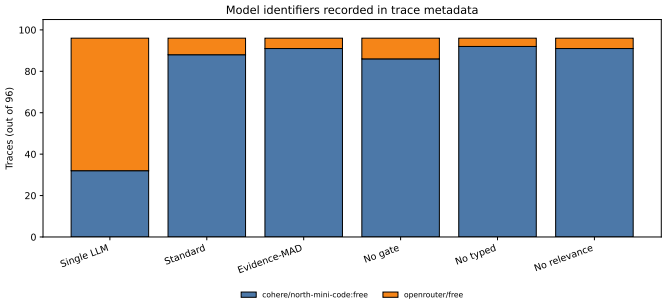


Fig. 8. Recorded OpenCode model identifiers per system. Counts sum to 96 in each bar.

when evidence is missing. It may also be failing to retrieve or normalize useful claims. The current data cannot distinguish those cases. Logging the number and quality of retrieved documents, then annotating whether they answer the query, would make the next study more informative. The traces also show that a completed evaluator record can hide a failed stage. The example trace for the Convention on the Rights of the Child query contains a factual conflict classification, but the proposer call failed and no debate round completed. The record still has a final answer. This example shows why completion, debate activation, completed debate, and answer correctness should be separate measures. The current summary reports call failures, but a future paper should also report the number of debates that reached each stage.

The model identifier mix is a further limitation. The six folders do not use one recorded alias in every trace, and the distribution differs by system. A controlled rerun should pin the same model identifier for all configurations and preserve an immutable run manifest. It should also use a stable retrieval snapshot or record the full retrieved documents so that external search changes do not affect comparisons.

VII. LIMITATIONS AND NEXT EVALUATION

This analysis has several limitations. First, the results are based on a single live run. Search rankings, provider behavior, network conditions, and model availability may change across repeated executions. Second, the available traces do not contain gold answer labels, so answer accuracy and conflict detection precision and recall cannot be reported. Third, system signatures vary by recorded OpenCode model identifier, making the comparisons observational. Fourth, trace completion does not guarantee successful completion of every model stage. Finally, evidence state labels are system outputs rather than independent annotations. Therefore, the figures describe system behavior but do not establish whether the decisions were correct.

Future evaluation should use labelled questions, preserve the retrieved evidence, and maintain a fixed model configuration. Human annotators should label answer facts, claim pair relations, and whether debate is necessary. The evaluation could then report answer accuracy, state confusion matrices, debate precision and recall, unnecessary debate rate, missed conflict rate, completed debate rate, calls, cost, and latency. Multiple runs would help assess the effects of live service variation. Until then, the current results should not be used to claim that Evidence MAD improves answer quality.

VIII. CONCLUSION

Evidence MAD combines retrieval, claim comparison, and conflict-aware debate. In 96 matched traces, it triggered debate on 22 questions, compared with 50 for Standard MAD. This is a useful description of routing behavior. It is not enough to show that the system is more accurate or that the gate selects the right questions. The traces have no gold answer labels, model identifiers vary across configurations, and some internal calls fail even when the evaluator saves a completed record. The next step is a controlled, labelled evaluation. The current traces provide a useful starting point: they show where the pipeline stops, how often it debates, and where latency accumulates. They do not yet establish that the proposed method performs better than the alternatives.

APPENDIX A
SAVED TRACE DATA

Each trace records the question, trace identifier, routing information, retrieved evidence, normalized claims, evidence decision, debate trigger, conflict metadata where available, debate rounds, final answer, timing and call metrics, and evaluator status. The JSON summary is recalculated from these trace level fields. The analysis also preserves the recorded model identifier counts to support interpretation of the results.

A. Project Resources

The following project materials are provided to support inspection and reproduction of the implementation and the reported trace analysis:

- *Code base (MAD 2):* https://github.com/sasikiran-workmode/MAD-2
- *Evaluation traces:* https://github.com/Suhas-Koheda/evidence-mad/tree/main/traces
- *Literature review:* https://github.com/Suhas-Koheda/evidence-mad/blob/main/Multi-Agent%20Debate%20Framework%20-%20Final%20Literature%20Review%20Updated.pptx
- *Architecture diagrams:* https://github.com/Suhas-Koheda/evidence-mad/tree/main/architecture

APPENDIX B
PRACTICAL READING OF THE FIGURES

The debate activation plot answers how often each trace records a debate trigger. It does not measure whether debate was warranted. Evidence state bars show the states recorded by each system; they are not a gold standard confusion matrix. Latency bars summarize one run and include the long tail of live retrieval and model calls. Call bars include successful and failed model calls across pipeline roles. Retrieval bars report trace metrics for usable agents and extracted claims. The model signature plot reports configured identifiers found in trace metadata, not a provider-side audit. The architecture figure summarizes intended control flow and is not a trace derived performance plot.

REFERENCES

[1] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W.-t. Yih, T. Rocktäschel, S. Riedel, and D. Kiela, "Retrieval-augmented generation for knowledge-intensive nlp tasks," *Advances in Neural Information Processing Systems*, vol. 33, pp. 9459–9474, 2020. [Online]. Available: https://arxiv.org/abs/2005.11401

[2] Y. Du, S. Li, A. Torralba, J. B. Tenenbaum, and I. Mordatch, "Improving factuality and reasoning in language models through multiagent debate," in *Proceedings of the 41st International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 235, 2024, pp. 11 733–11 763. [Online]. Available: https://proceedings.mlr.press/v235/du24e.html

[3] J. C.-Y. Chen, S. Saha, and M. Bansal, "Reconcile: Round-table conference improves reasoning via consensus among diverse llms," in *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Association for Computational Linguistics, 2024, pp. 7066–7085. [Online]. Available: https://aclanthology.org/2024.acl-long.381/

[4] T. Liang, Z. He, W. Jiao, X. Wang, Y. Wang, R. Wang, Y. Yang, S. Shi, and Z. Tu, "Encouraging divergent thinking in large language models through multi-agent debate," in *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, 2024, pp. 17 889–17 904. [Online]. Available: https://aclanthology.org/2024.emnlp-main.992/

[5] C.-M. Chan, W. Chen, Y. Su, J. Yu, W. Xue, S. Zhang, J. Fu, and Z. Liu, "Chateval: Towards better llm-based evaluators through multi-agent debate," in *The Twelfth International Conference on Learning Representations*, 2024. [Online]. Available: https://proceedings.iclr.cc/paper_files/paper/2024/hash/25cc3adf8c85f7c70989cb8a97a691a7-Abstract-Conference.html

[6] A. Khan, J. Hughes, D. Valentine, L. Ruis, K. Sachan, A. Radhakrishnan, E. Grefenstette, S. R. Bowman, T. Rocktäschel, and E. Perez, "Debating with more persuasive LLMs leads to more truthful answers," in *Proceedings of the 41st International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 235, 2024, pp. 23 662–23 733. [Online]. Available: https://proceedings.mlr.press/v235/khan24a.html

[7] K. Xiong, X. Ding, Y. Cao, T. Liu, and B. Qin, "Examining inter-consistency of large language models collaboration: An in-depth analysis via debate," in *Findings of the Association for Computational Linguistics: EMNLP 2023*. Association for Computational Linguistics, 2023, pp. 7572–7590. [Online]. Available: https://aclanthology.org/2023.findings-emnlp.508/

[8] S. Eo, H. Moon, E. H. Zi, C. Park, and H. Lim, "Debate only when necessary: Adaptive multiagent collaboration for efficient llm reasoning," 2025. [Online]. Available: https://arxiv.org/abs/2504.05047

[9] J. Baba and J. A. Chudziak, "From argument components to graphs: A multi-agent debate with confidence gating for argument relations," 2026, preprint. [Online]. Available: https://arxiv.org/abs/2606.16047

[10] H. K. Choi, X. Zhu, and Y. Li, "Debate or vote: Which yields better decisions in multi-agent large language models?" in *Advances in Neural Information Processing Systems*, 2025. [Online]. Available: https://arxiv.org/abs/2508.17536

[11] H. Wang, X. Du, W. Yu, Q. Chen, K. Zhu, Z. Chu, L. Yan, and Y. Guan, "Learning to break: Knowledge-enhanced reasoning in multi-agent debate system," *Neurocomputing*, vol. 618, p. 129063, 2025.

[12] C.-Y. Chang, Z. Jiang, V. Rakesh, M. Pan, C.-C. M. Yeh, G. Wang, M. Hu, Z. Xu, Y. Zheng, M. Das, and N. Zou, "MAIN-RAG: Multi-agent filtering retrieval-augmented generation," in *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Association for Computational Linguistics, 2025, pp. 2607–2622. [Online]. Available: https://aclanthology.org/2025.acl-long.131/

[13] H. Wang, A. Prasad, E. Stengel-Eskin, and M. Bansal, "Retrieval-augmented generation with conflicting evidence," in *Proceedings of the Conference on Language Modeling*, 2025. [Online]. Available: https://openreview.net/forum?id=z1MHB2m3V9

[14] S. Jeong, Y. Choi, J. Kim, and B. Jang, "Tool-MAD: A multi-agent debate framework for fact verification with diverse tool augmentation and adaptive retrieval," 2026, preprint. [Online]. Available: https://arxiv.org/abs/2601.04742

[15] A. Cattan, A. Jacovi, O. Ram, J. Herzig, R. Aharoni, S. Goldshtein, E. Ofek, I. Szpektor, and A. Caciularu, "Dragged into conflicts: Detecting and addressing conflicting sources in search-augmented llms," 2025, preprint. [Online]. Available: https://arxiv.org/abs/2506.08500

[16] R. Xu, Z. Qi, Z. Guo, C. Wang, H. Wang, Y. Zhang, and W. Xu, "Knowledge conflicts for llms: A survey," in *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, 2024, pp. 8541–8565. [Online]. Available: https://aclanthology.org/2024.emnlp-main.486/

[17] K. Huang, Y. Cai, X. Wu, Z. Deng, and L. Yuan, "SMADE-IE: Sparse multi-agent framework with evidence-driven debate for zero-shot information extraction," 2026, preprint. [Online]. Available: https://arxiv.org/abs/2606.04691

[18] A. Williams, N. Nangia, and S. R. Bowman, "A broad-coverage challenge corpus for sentence understanding through inference," in *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, 2018, pp. 1112–1122. [Online]. Available: https://aclanthology.org/N18-1101/